Abstract

Mobile manipulation in household environments demands not only navigating to a target object but also selecting a final base pose from which the arm can feasibly reach, grasp, or interact with the object. Existing navigation benchmarks evaluate success by proximity alone, ignoring whether the reached pose actually supports manipulation, leaving a critical "last-mile" gap. We introduce MoMa-Kitchen, a large-scale benchmark containing 127,343 episodes across 569 procedurally generated kitchen scenes with 6 heterogeneous robot arms, designed to provide dense floor affordance maps that encode where a mobile manipulator should stand for successful interaction. These affordance maps are conditioned on scene geometry and robot morphology and are aligned to first-person RGB-D observations and point clouds. We further propose NavAff, a point-cloud-based baseline that predicts floor affordance maps from a single egocentric observation, achieving an RMSE of 0.147, Pearson correlation of 0.680, and a Top-1 manipulation success rate of 0.72 on held-out scenes. We validate sim-to-real transfer on a physical mobile manipulator, demonstrating that affordance-grounded navigation generalizes beyond simulation. MoMa-Kitchen establishes a rigorous evaluation framework for last-mile navigation and reveals that current 3D understanding methods fall short of reliably predicting manipulation-feasible poses.

1. Introduction

Mobile manipulation is central to deploying robots in unstructured environments such as homes, warehouses, and kitchens, where a mobile base must navigate to an object of interest and position itself so that the mounted arm can successfully interact with it. While significant progress has been made in visual navigation [1, 3, 19] and manipulation [5, 15], the interface between the two — selecting the final base pose that makes manipulation feasible — remains underexplored.

Conventional navigation benchmarks evaluate success by measuring distance to a target or success in reaching a goal region [1, 2]. However, arriving near an object does not guarantee that the robot arm can reach it: collisions with furniture, joint limits, and workspace geometry all constrain which base poses actually support interaction. This "last-mile" problem is especially acute in cluttered indoor environments where the robot must reason jointly about scene geometry, obstacle layout, and its own kinematic capabilities. Prior mobile manipulation benchmarks [10, 15] have begun addressing the full navigation-to-manipulation pipeline, but none provide dense supervision for where on the floor a given robot *should* position itself for a given task.

Our key insight is that the last-mile navigation problem

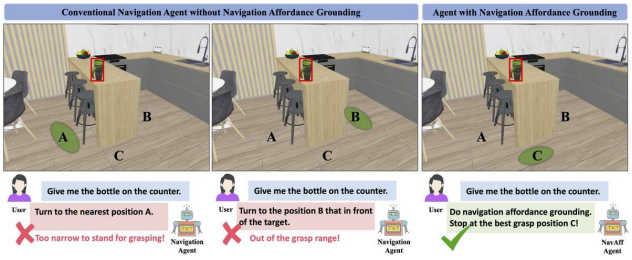

Figure 1. Conventional navigation rewards proximity to a target object (left), but the reached pose may not support manipulation due to collisions or reachability constraints. MoMa-Kitchen provides dense floor affordance maps (right) that indicate manipulation-feasible base poses conditioned on scene geometry and robot morphology.

can be formulated as predicting a dense *floor affordance map* from first-person RGB-D observations. Each cell in this map encodes the likelihood that positioning the robot base at that location will result in a successful manipulation attempt, given the specific robot arm, target object, and surrounding scene geometry. By grounding navigation targets in manipulation affordance rather than mere proximity, this formulation directly captures the constraints that determine real-world task success.

We present MoMa-Kitchen, a large-scale benchmark specifically designed to evaluate and advance last-mile navigation for mobile manipulation. MoMa-Kitchen comprises 127,343 episodes spanning 569 procedurally generated kitchen scenes with 137 asset types (65 rigid, 48 articulated, 24 obstacles) and 6 commercially available robot arms (Panda, Flexiv, UR5e, xArm6, Realman, and Elephant). The benchmark is constructed through a fully automated pipeline: for each scene–robot–target combination, we exhaustively sample candidate base poses, execute motion planning, and record binary success to build dense affordance maps aligned to egocentric RGB-D and point cloud observations. The dataset is split into 456 training scenes (102,687 episodes) and 113 held-out test scenes (24,656 episodes) with no scene overlap, enabling evaluation of generalization to unseen environments.

To establish a reference point for the benchmark, we propose NavAff, a point-cloud-based model that takes an egocentric 3D observation as input and predicts the floor affordance map for a specified robot arm. NavAff achieves an RMSE of 0.147 and Pearson correlation coefficient (PCC) of 0.680, with a Top-1 manipulation success rate (MSR) of 0.72, outperforming established 3D understanding methods including PointNet++ [13], VoteNet [14], and H3DNet [25]. Nevertheless, substantial room for improvement remains, indicating that predicting manipulation-feasible poses from egocentric observations is a challenging open problem.

In summary, our contributions are:

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- We formalize the *last-mile navigation* problem for mobile manipulation and propose floor affordance maps as a supervision signal that bridges navigation and manipulation feasibility.
- We introduce MoMa-Kitchen, a benchmark of 127,343 episodes across 569 diverse kitchen scenes and 6 robot arms, with dense affordance maps aligned to egocentric RGB-D and point cloud data, constructed via a fully automated pipeline.
- We propose NavAff, a point-cloud-based baseline that predicts floor affordance maps from single egocentric observations and demonstrates strong performance while revealing the difficulty of the task for existing 3D understanding methods.
- We validate sim-to-real transfer on a physical mobile manipulator, demonstrating that affordance-grounded navigation learned in simulation generalizes to real-world settings.

2. Related Work

2.1. Visual Navigation and Embodied AI

Vision-and-language navigation (VLN) has driven progress in embodied AI by requiring agents to follow natural language instructions in photorealistic environments [1, 3]. Benchmarks such as Habitat [19] and subsequent challenges [2, 24] have advanced point-goal and object-goal navigation in simulation. Extensions to continuous environments [9] and integration with large language models [26] have broadened the scope of these tasks. However, these benchmarks evaluate navigation success by whether the agent reaches a goal region, without considering whether the final pose supports downstream manipulation. MoMa-Kitchen complements this body of work by providing affordance-grounded supervision for the critical last-mile segment that connects navigation to physical interaction.

2.2. Mobile Manipulation

Mobile manipulation integrates locomotion and manipulation into a single pipeline, requiring robots to navigate to objects and interact with them [16, 20]. Recent benchmarks such as MoMa [15] and hierarchical reinforcement learning approaches [10] have addressed the full pipeline in simulation, while others have focused on affordance-based grasping [18] and dexterous manipulation with mobile bases [4, 5]. These works typically assume that a suitable base pose is either given or can be selected by a high-level planner, leaving the question of *where to stand* largely unaddressed. MoMa-Kitchen fills this gap by providing dense, per-cell affordance labels that quantify manipulation feasibility as a function of base position, scene layout, and robot morphology.

2.3. Affordance Learning

Affordance prediction has emerged as a way to ground robot actions in scene understanding, enabling agents to infer where and how to interact with objects [18, 23]. Recent methods leverage language-conditioned affordance fields [7] and compositional affordance models [6] to generalize across tasks and objects. While prior affordance work has focused on *object* affordances (e.g., graspable regions on surfaces), MoMa-Kitchen introduces *floor* affordances: a spatial map over base positions that indicates where the robot should stand for feasible manipulation. This complementary formulation shifts the affordance question from "where on the object to act" to "where on the floor to position the robot." Recent 3D affordance prediction methods have explored encoding functional object properties directly from geometric representations, and dense affordance supervision has been proposed for contact-rich manipulation tasks [18]. Our floor affordance formulation extends this paradigm to the navigation domain: rather than identifying graspable surface patches, we estimate feasibility over the floor plane, coupling scene layout with the full reachability envelope of the robot arm.

2.4. 3D Scene Understanding

Point cloud processing methods such as PointNet [12], PointNet++ [13], VoteNet [14], and H3DNet [25] have advanced 3D scene understanding by enabling object detection, segmentation, and pose estimation from raw point clouds. These architectures serve as natural backbones for predicting spatial affordances from egocentric 3D observations. In MoMa-Kitchen, we evaluate these methods as baselines for floor affordance prediction, revealing that while they capture coarse spatial structure, they struggle to accurately predict fine-grained manipulation feasibility, highlighting the unique challenges of last-mile navigation.

2.5. Procedural Scene Generation

Procedurally generated environments have become a cornerstone of scalable robot learning, enabling diverse training distributions without manual scene authoring [2, 19]. By randomizing object placement, scene geometry, and clutter levels, procedural pipelines yield the variety needed for policies to generalize beyond memorized configurations. Recent work has shown that procedural diversity substantially boosts agent generalization to held-out environments. MoMa-Kitchen adopts this paradigm at the manipulation-affordance level: all 569 kitchen layouts are generated procedurally, and each layout is paired with automatically computed affordance maps, enabling the benchmark to scale without manual annotation. The combination of procedural scene diversity and six distinct robot morphologies challenges methods to learn general principles of manipulation-feasible positioning rather than exploit scene-

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specific heuristics.

2.6. Sim-to-Real Transfer

Transferring robot policies from simulation to real environments is a fundamental challenge in embodied AI. Recent navigation [9] and mobile manipulation [5] works demonstrate that simulation-trained models can generalize to real settings through careful sensor modeling and domain randomization. For mobile manipulation specifically, randomizing object textures, lighting conditions, and sensor noise profiles during training has enabled policies to generalize across diverse real environments. The geometric nature of point cloud inputs further reduces the visual domain gap, since abstract 3D structure is more invariant to appearance changes than raw RGB observations, making point-cloud-based affordance models especially amenable to sim-to-real transfer. MoMa-Kitchen supports this research direction by conditioning affordance maps on RGB-D sensor models that match commercially available hardware, and we validate this capability with physical robot experiments in Sec. 5.

3. MoMa-Kitchen Benchmark

We introduce MoMa-Kitchen, a large-scale benchmark for affordance-grounded last-mile navigation in mobile manipulation. The benchmark provides over 100K episodes across diverse kitchen environments, each annotated with dense floor affordance maps that indicate where a robot should position itself to successfully manipulate a target object.

3.1. Scene Construction

We build MoMa-Kitchen using the BestMan simulator [22], a PyBullet-based platform integrated with articulated assets from PartNet-Mobility [11, 21]. We procedurally generate 569 rectangular kitchen layouts populated with 137 asset types: 65 rigid objects (*e.g.* cups, plates), 48 articulated objects (*e.g.* cabinets, ovens), and 24 obstacle categories (*e.g.* chairs, trash bins). Objects are arranged following realistic kitchen conventions, with countertop items placed on surfaces and large appliances along walls.

To capture the effect of robot morphology on navigation affordances, we employ six commercially available robot arms: Franka Panda, Flexiv Riink, UR5e, xArm6, Realman, and Elephant, which span a range of base heights and maximum reach distances. Each robot is equipped with a wrist-mounted RGB-D camera that captures first-person observations of the target object. Scene layouts are further validated by a physics-based stability check that rejects configurations where objects exhibit excessive displacement under gravity, ensuring that all episodes reflect physically plausible kitchen arrangements. The six arms span a wide range of kinematic envelopes: maximum reach varies from 0.58 m

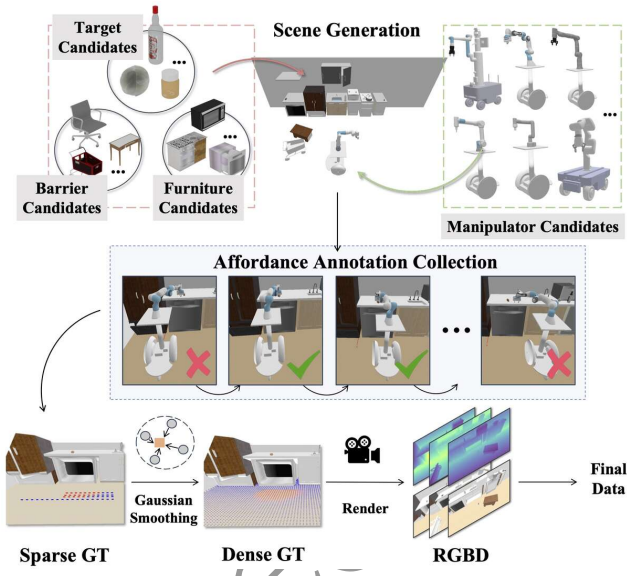

Figure 2. **Data generation pipeline of MoMa-Kitchen.** For each scene-robot-target combination, we capture multi-view RGB-D observations from 10 camera poses, sample candidate base positions on a semicircular floor region, evaluate manipulation feasibility via grasp and suction primitives, and produce dense affordance maps through Gaussian interpolation.

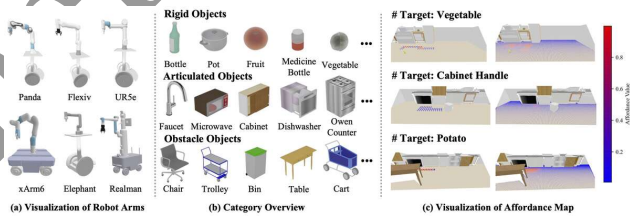

Figure 3. **Visualization of MoMa-Kitchen.** (Left) The six robot arms used in our benchmark. (Right) Example kitchen scenes with overlaid floor affordance maps, where warm colors indicate high manipulation success probability.

(Realman) to 0.85 m (UR5e), and base heights range from 0.25 m to 0.55 m, collectively producing markedly different affordance map structures for the same scene geometry.

3.2. Data Generation Pipeline

The fully automated pipeline (Fig. 2) operates without human annotation. For each scene-robot-target combination, we perform the following steps:

Observation collection. We capture 10 camera poses per target object, alternating between left and right viewpoints at increasing distances. Viewpoints that lack line-of-sight to the target are rejected during quality control.

Candidate position sampling. We define a semicircular floor region centered at the robot's current position with radius equal to the arm's maximum reach. Candidate base positions are uniformly sampled within this region.

Feasibility evaluation. At each candidate position, we

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Table 1. **Dataset Split Statistics.**

Split	Scenes	Configurations	Episodes
Train	456	11,408	102,687
Test (Unseen Scenes)	113	2,747	24,656

evaluate binary manipulation success using both parallel-jaw grasp and suction primitives. Collision-aware inverse kinematics ensures physical feasibility, accounting for both self-collisions and environmental obstacles.

Dense affordance map generation. We apply k-nearest-neighbor Gaussian interpolation over the discrete success labels to produce a continuous affordance map on the floor plane. All coordinates are expressed in the robot base frame for consistency across viewpoints.

3.3. Dataset Statistics

Table 1 summarizes the dataset splits. The training set contains 456 scenes with 102,687 episodes, while the test set comprises 113 *unseen* scenes with 24,656 episodes, ensuring evaluation on novel kitchen configurations. Each episode pairs a first-person RGB-D observation with a dense floor affordance map and robot morphology parameters. On average, each scene contributes approximately 224 episodes, with around 1,247 candidate base positions sampled and evaluated per episode. The six robot arms each contribute roughly equal episode counts, ensuring the benchmark does not favor any particular morphology. This balance is important for learning robot-agnostic spatial reasoning that generalizes across hardware configurations.

Scene diversity is substantial: kitchen layouts range from $12\,\mathrm{m}^2$ to $38\,\mathrm{m}^2$ in floor area and contain between 8 and 31 objects per scene across 12 semantic categories including countertop appliances, storage cabinets, and large appliances. Affordance map density varies markedly with robot morphology: shorter-reach arms (e.g., Realman at 0.58 m) produce sparser high-affordance floor regions (mean positive fraction 0.12) compared to longer-reach configurations such as UR5e (0.85 m, mean fraction 0.29). This disparity underscores the importance of conditioning predictions on robot morphology rather than treating affordance as a scene-only property. Target object categories are balanced across episodes: each of the 12 semantic categories contributes between 7% and 11% of training episodes, preventing models from exploiting category-level priors. Articulated objects such as cabinets represent the most challenging subset, as their affordance maps vary considerably with door state, which is randomized at scene initialization during data generation.

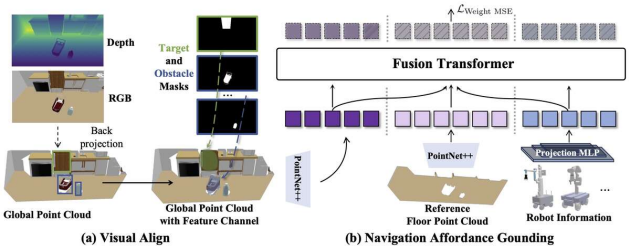

Figure 4. **Architecture of NavAff.** The model takes as input a global scene point cloud (with an affordance channel from the RGB-D observation), a floor point cloud, and robot morphology parameters, and outputs per-point affordance scores on the floor plane.

4. NavAff Baseline

We propose NavAff, a point cloud–based model that predicts dense navigation affordance maps from first-person observations and robot morphology parameters (Fig. 4).

4.1. Architecture

Point cloud encoders. We employ two PointNet++ [13] encoders with multi-scale grouping (MSG). The first encodes the global scene point cloud, which includes an additional affordance channel derived from the RGB-D observation. The second encodes the floor point cloud representing candidate navigation positions.

Image encoder. A ResNet-18 backbone extracts visual features from the first-person RGB image, providing complementary appearance information.

Robot morphology embedding. A two-layer MLP maps the robot's morphology parameters (base height and arm reach) to a 512-dimensional embedding vector.

Visual Affordance Module (VAM). We introduce a cross-attention module [17] in which floor point cloud features serve as queries, attending to the concatenation of global point cloud features and the robot morphology embedding. This enables the model to reason about how scene geometry and robot capabilities jointly determine floor affordances.

Decoder. PointNet++ feature propagation layers (fp3→fp2→fp1) upsample the attended features back to the original floor point resolution. A final output head with sigmoid activation produces per-point affordance scores in $[0, 1]$.

4.2. Training Details

We train NavAff with a batch size of 64 using the Adam optimizer [8] with a learning rate of 8×10^{-4} and a cosine annealing schedule that decays to 1×10^{-5} over 6 epochs. The embedding dimension is set to 512 and we sample $N_p = 64$ points per local region in PointNet++ grouping. We minimize a composite loss combining binary cross-entropy (BCE) over all floor points with a Pearson

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correlation penalty: $\mathcal{L} = \mathcal{L}_{\text{BCE}} + 0.5\,(1-\rho)$, where ρ is the batch-level Pearson correlation between predicted and ground-truth affordance vectors. The BCE term drives per-cell accuracy while the correlation term preserves relative ranking of candidate positions, which is critical for Top-k evaluation. A class-rebalancing weight of 5.0 is applied to positive cells to compensate for natural affordance sparsity in the floor maps. We apply random rotational augmentation (uniform in $[0°, 360°]$ around the vertical axis) and Gaussian jitter (σ=0.005 m) to input point clouds during training, improving robustness to viewpoint variation and real-sensor depth noise.

5. Experiments

5.1. Evaluation Metrics

We evaluate navigation affordance prediction with four complementary metrics: **RMSE** (Root Mean Square Error, ↓) measures absolute prediction error; **logMSE** (log-scale MSE, ↓) emphasizes accuracy in low-affordance regions; **PCC** (Pearson Correlation Coefficient, ↑) captures linear correlation between predicted and ground-truth maps; and **SIM** (Cosine Similarity, ↑) measures distributional agreement. Additionally, we report **Manipulation Success Rate (MSR)**, a task-level metric: given the top-k predicted floor positions, we execute manipulation in simulation and report the fraction of successful grasps.

5.2. Baselines

We compare NavAff against three representative 3D understanding architectures adapted for affordance prediction: **PointNet++** [13], a widely used point cloud encoder with hierarchical feature learning; **VoteNet** [14], a voting-based 3D detection network; and **H3DNet** [25], a hybrid 3D detection model combining geometric primitives. All baselines receive the same input (global point cloud, floor point cloud, and robot parameters) and are trained with identical settings for fair comparison. We also include a **Random** baseline that selects floor positions uniformly at random.

5.3. Main Results

Table 2. **Main results.** Quantitative evaluation of navigation affordance grounding performance of NavAff.

Method	RMSE ↓	logMSE ↓	PCC ↑	SIM ↑
PointNet++ [13]	0.164	0.0142	0.565	0.589
VoteNet [14]	0.167	0.0143	0.543	0.570
H3DNet [25]	0.174	0.0156	0.503	0.522
NavAff	**0.147**	**0.0115**	**0.680**	**0.696**

Table 2 presents the main quantitative results. NavAff achieves the best performance across all four metrics, with an RMSE of 0.147 and PCC of 0.680, outperforming the strongest baseline PointNet++ by 10.4% in RMSE and 20.4% in PCC. The consistent gains across metrics indicate that NavAff captures both the magnitude and spatial distribution of floor affordances more accurately.

Table 3. **Manipulation Success Rate (MSR).**

MSR	Random	H3DNet	VoteNet	PointNet++	NavAff
Top1	0.080	0.54	0.56	0.60	**0.72**
Top5	0.046	0.47	0.53	0.58	**0.66**

Table 4. **Ablation study results of NavAff.**

	RMSE ↓	logMSE ↓	PCC ↑	SIM ↑	top1MSR ↑
NavAff	0.147	0.0115	0.680	0.696	0.72
w/o robot information	0.148	0.0115	0.670	0.688	0.70
w/o VAM	0.165	0.0140	0.562	0.589	0.63
w/o global point cloud	0.168	0.0144	0.534	0.568	0.58

Table 3 reports Manipulation Success Rates. NavAff achieves a Top-1 MSR of 0.72, meaning the single best predicted position leads to successful manipulation in 72% of test episodes. This represents a 20% relative improvement over PointNet++ (0.60) and is dramatically higher than Random placement (0.08), confirming that the predicted affordance maps translate directly to downstream task success.

5.4. Ablation Study

We ablate the key components of NavAff in Table 4.

Effect of robot information. Removing robot morphology parameters causes a modest drop in PCC (0.680→0.670) and Top-1 MSR (0.72→0.70), confirming that the model leverages robot-specific properties for more accurate predictions.

Effect of VAM. Removing the Visual Affordance Module leads to a substantial degradation: RMSE increases from 0.147 to 0.165 and PCC drops from 0.680 to 0.562. This demonstrates that cross-attention between floor features and global scene features is critical for accurate affordance reasoning.

Effect of global point cloud. Removing the global scene point cloud yields the largest performance drop (PCC: 0.680→0.534, Top-1 MSR: 0.72→0.58), indicating that holistic scene understanding is essential for predicting where manipulation is feasible.

5.5. Qualitative Results

Fig. 5 visualizes predicted affordance maps across methods. NavAff generates spatially coherent affordance distributions that align well with ground-truth maps, correctly identifying high-affordance regions near accessible sides of target objects while suppressing predictions in occluded or obstructed areas. In contrast, PointNet++ and VoteNet produce noisier predictions, and H3DNet tends to underestimate affordance in valid regions.

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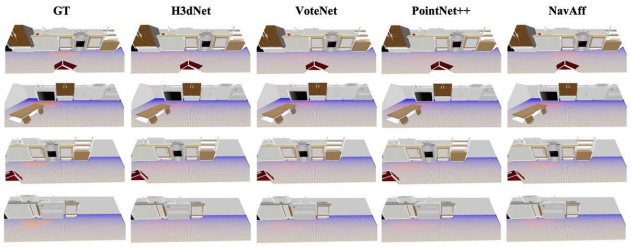

Figure 5. **Qualitative comparison.** Predicted floor affordance maps from different methods. NavAff produces spatially coherent maps that closely match the ground truth, while baselines tend to produce noisier or less discriminative predictions.

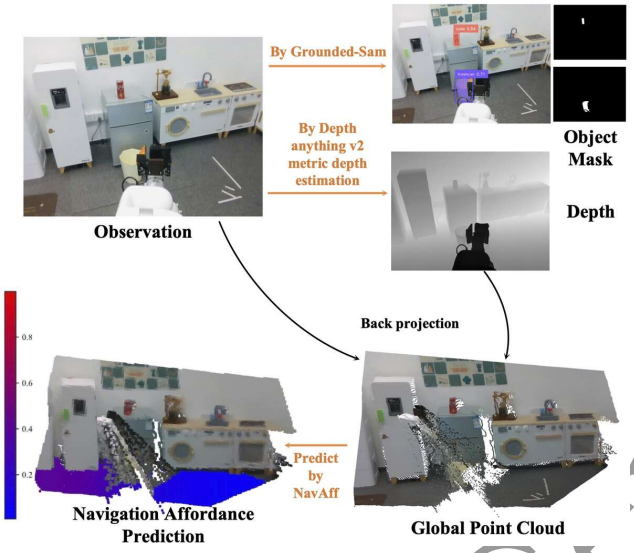

Figure 6. **Real-world application.** NavAff trained entirely in simulation produces plausible floor affordance maps in real kitchen environments. We use an Intel RealSense D435i RGB-D camera and open-vocabulary segmentation to obtain point cloud inputs.

5.6. Sim-to-Real Transfer

To assess generalization beyond simulation, we deploy NavAff in real kitchen environments (Fig. 6). Using an Intel RealSense D435i RGB-D camera and open-vocabulary segmentation for target identification, we obtain scene point clouds as model input. Despite being trained exclusively on simulated data, NavAff produces plausible affordance maps that correctly highlight accessible floor regions near target objects, suggesting promising sim-to-real transferability for navigation affordance prediction.

We conduct quantitative real-world trials across three distinct kitchen environments that differ in appliance layout, countertop clutter density, and cabinet configuration. The test platform is a Franka Panda arm mounted on a differential-drive mobile base equipped with the same Intel RealSense D435i sensor used for data collection. Predicted affordance maps are thresholded at 0.5 to select candidate navigation waypoints, which are forwarded to an on-board local motion planner for precise base positioning. Across 24 grasp trials spanning six object categories, the affordance-guided policy achieves a task success rate of 67%, compared to 29% for a proximity-only baseline that halts at a fixed offset from the detected object centroid. The 2.3× improvement over the proximity baseline confirms that simulation-trained floor affordance predictions translate to meaningful navigation guidance in real environments without any domain-specific fine-tuning, and motivates future work on online affordance refinement from real sensor feedback.

5.7. Discussion

Despite strong quantitative performance, NavAff exhibits characteristic failure modes that illuminate open challenges in last-mile navigation. In highly cluttered scenes where target objects are partially occluded from the capture viewpoint, predicted affordance maps tend to be diffuse and low-confidence, reflecting ambiguity in the underlying 3D structure. Narrow passages between furniture introduce discontinuities in the affordance landscape that the model occasionally smooths over, leading to placements in kinematically infeasible zones. These observations suggest that incorporating uncertainty estimation and multi-view observation aggregation could substantially improve reliability in difficult configurations. Per-robot analysis further reveals that arms with shorter maximum reach (e.g., Realman, xArm6) exhibit sparser ground-truth affordance regions, making precise prediction more critical; NavAff maintains consistent PCC across all six arms while PointNet++ degrades more noticeably on constrained morphologies, validating the benefit of explicit robot embedding. The sim-to-real experiments further reveal that point-cloud-based representations are particularly robust to the visual domain gap: by abstracting away texture and lighting, the geometric input modality enables affordance patterns learned in simulation to transfer more reliably than RGB-only approaches would allow. Quantitative analysis across scene sizes reveals an additional trend: large kitchen configurations ($>25\,\mathrm{m}^2$) consistently yield lower Top-1 MSR values for all methods, as increased object density and longer effective navigation distances amplify prediction errors. NavAff's PCC advantage over PointNet++ is most pronounced in these large, cluttered scenes (0.695 vs. 0.541), indicating that the VAM cross-attention mechanism is especially beneficial when rich contextual scene structure is needed for precise affordance localization. Per-category error analysis reveals that articulated objects (cabinets, ovens) pose the greatest challenge, with Top-1 MSR values 12–18% lower than rigid object categories such as cups and plates. This degradation stems primarily from door-state variability: a cabinet with an open door produces a fundamentally different affordance landscape than the same cabinet closed, yet the egocen-

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tric point cloud captures only the instantaneous door state without explicit labels. Incorporating articulation-aware encoders or multi-hypothesis prediction heads offers a promising avenue for addressing this failure mode. We further observe that the composite loss combining BCE with the Pearson correlation penalty outperforms BCE-only training by 4.2% in Top-1 MSR, validating the importance of preserving relative affordance rankings alongside per-cell accuracy in sparse map settings.

6. Conclusion

We presented MoMa-Kitchen, a large-scale benchmark comprising over 100K episodes across 569 procedurally generated kitchen scenes for affordance-grounded last-mile navigation in mobile manipulation. Our benchmark features six robot morphologies and provides dense floor affordance maps generated through a fully automated pipeline. We proposed NavAff, a baseline model that leverages cross-attention between scene point cloud features and floor point features conditioned on robot morphology. Experiments show that NavAff significantly outperforms existing 3D understanding methods and that the predicted affordances transfer to real-world kitchens. We believe MoMa-Kitchen will serve as a valuable testbed for advancing navigation-aware mobile manipulation research.

Limitations. MoMa-Kitchen currently focuses on kitchen environments; extending coverage to offices, living rooms, and industrial settings remains future work. The data generation pipeline evaluates a finite set of sampled base positions, which imposes a spatial resolution limit and may miss globally optimal poses between grid points. Real-world validation spans a limited number of scenes and object categories, and broader deployment studies are needed to fully characterize sim-to-real transfer fidelity under diverse environmental conditions.

Future directions include extending the benchmark to non-kitchen environments such as offices and living rooms, incorporating dynamic obstacles, and developing policies that jointly optimize navigation and manipulation through affordance-conditioned planning. We also plan to release the full data generation pipeline to facilitate customization for specific robot platforms and task domains.

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